Distributed Inference for Extreme Expectiles of Heavy-Tailed Data: Optimal Pooling of Hill and Quantile Statistics

Setting. Distributed (one-shot, communication-constrained) inference with a *fixed* number m of machines holding independent and identically distributed data, as in Section 3 of Daouia, Padoan and Stupfler (2024) — hereafter **DPS**. Each machine communicates the statistics prescribed by the DPS heuristics: its subsample size, its effective sample size, its Hill estimator and its intermediate order statistic (plus, optionally, estimators of the second-order parameters for bias-aware procedures). **Target: an extreme expectile** ξ_{1-p} with $p = p(n) \downarrow 0$, possibly $np = O(1)$.

Scope of the proofs. All results that are *new* (everything concerning expectiles in the distributed setting: Lemma 1, Theorems 1, 2, 5(i)–(iii), 6, Propositions 3, 10 and Corollaries 4, 7, 8, 9) are stated and **proved in full**. Results imported from the three reference papers and from classical extreme value theory are stated precisely as Propositions A–E with exact citations and are *not* reproved; this is the standard division of labour in a thesis chapter. No finite-sample simulation study is developed, as requested.

1. The problem

A sample of n observations from a heavy-tailed distribution is scattered across m machines (data centres, insurance branches, hospitals, …), machine j holding n_j observations

$$X_{1,j}, \dots, X_{n_j,j} \overset{\text{i.i.d.}}{\sim} F, \qquad 1 \le j \le m, \qquad n = \sum_{j=1}^{m} n_j,$$

with full independence within and across machines and a *common* distribution function F. Raw data cannot be shared (privacy, bandwidth, storage). Following the communication heuristics of DPS (Sections 2.2–2.3 and 3 therein), machine j transmits to the central machine **only** the vector

$$T_j \;=\; \Big(n_j,\ k_j,\ \widehat{\gamma}_j(k_j),\ X_{n_j-k_j:n_j,j}\Big), \qquad \text{optionally augmented by} \qquad \big(\widehat{\beta}_j, \widehat{\rho}_j\big),$$

where $X_{1:n_j,j} \le \cdots \le X_{n_j:n_j,j}$ are the order statistics of the j-th subsample,

$$\widehat{\gamma}_j(k_j) \;=\; \frac{1}{k_j} \sum_{i=1}^{k_j} \log \frac{X_{n_j-i+1:n_j,j}}{X_{n_j-k_j:n_j,j}}$$

is the Hill (1975) estimator of the tail index computed on the top k_j observations of machine j, $X_{n_j-k_j:n_j,j}$ is the corresponding intermediate order statistic, and $(\widehat{\beta}_j, \widehat{\rho}_j)$ are estimators of the second-order parameters (e.g. those of Gomes and Martins, as implemented in the `evt0` routines used by DPS).

The question. *How should the central machine pool the T_j to estimate the extreme expectile ξ_{1-p}, and with which asymptotic guarantees and optimal weights?*

Recall (Newey and Powell, 1987) that the expectile of F at level $\tau \in (0,1)$ is

$$\xi_\tau \;=\; \arg\min_{\theta \in \mathbb{R}} \mathbb{E}\big[\eta_\tau(X-\theta) - \eta_\tau(X)\big], \qquad \eta_\tau(u) = \big|\tau - 1\{u \le 0\}\big|\, u^2,$$

which is well defined as soon as $\mathbb{E}|X| < \infty$, and is the unique solution of the first-order condition

$$\tau \;=\; \frac{\mathbb{E}\big[|X-\xi_\tau|\, 1\{X \le \xi_\tau\}\big]}{\mathbb{E}|X-\xi_\tau|}.$$

Expectiles are the only coherent law-invariant elicitable risk measures besides the mean, and unlike quantiles they depend on the whole tail; this motivates their use for tail risk assessment (Davison, Padoan and Stupfler, 2023; Padoan and Stupfler, 2022).

Two obstacles distinguish this problem from the extreme quantile problem solved by DPS:

1. **No expectile information is transmitted.** The protocol shares tail-index and quantile information only. Any asymmetric-least-squares (LAWS) estimation of expectiles requires access to the raw subsamples and is therefore *infeasible* under the protocol. The solution must be **quantile-based (QB)**: it must convert pooled quantile/tail-index information into expectile information.
2. **A genuine extreme level.** The target level $1-p$ satisfies $np = O(1)$ in the leading case, so ξ_{1-p} lies beyond the range of every subsample and extrapolation is unavoidable; on top of the usual Weissman (1978) extrapolation error, the expectile–quantile relationship contributes additional bias terms that must be controlled.

The solution in one sentence. By the Bellini–Klar–Müller–Rosazza Gianin (2014) asymptotic proportionality $\xi_\tau/q_\tau \to (\gamma^{-1}-1)^{-\gamma}$ as $\tau \uparrow 1$, each machine's transmitted pair $(\widehat{\gamma}_j, X_{n_j-k_j:n_j,j})$ determines a Weissman-type extreme *expectile* estimator; pooling these **geometrically** with weights ω produces an estimator whose $\sqrt{k}/\log(k/(np))$-asymptotics are *entirely driven by the weighted pooled Hill estimator* $\widehat{\gamma}_n(\omega) = \sum_j \omega_j \widehat{\gamma}_j$, so that the whole DPS optimal-weighting theory (variance-optimal weights, oracle property, AMSE-optimal weights, bias reduction) transfers to extreme expectiles — with complete proofs given below.

2. Framework, communication protocol and assumptions

2.1 Notation

Throughout, $\overline{F} = 1-F$ denotes the survival function, $U(t) = \inf\{x : \overline{F}(x) \le 1/t\} = q_{1-1/t}$ ($t>1$) the tail quantile function, and $q_\tau = U(1/(1-\tau))$ the quantile of level τ. We write

$$\varphi(s) \;:=\; (s^{-1}-1)^{-s}, \qquad m(s) \;:=\; \frac{\mathrm{d}}{\mathrm{d}s}\log\varphi(s) \;=\; \frac{1}{1-s} - \log(s^{-1}-1), \qquad s\in(0,1),$$

for the *Bellini proportionality factor* and its log-derivative (the function m is precisely the one appearing in the asymptotics of QB expectile estimators in Davison, Padoan and Stupfler, 2023, and Padoan and Stupfler, 2022). We abbreviate

$$d_n := \frac{k}{np}, \quad D_n := \log d_n, \qquad d_{n,j} := \frac{k_j}{n_j p}, \quad D_{n,j} := \log d_{n,j},$$

and write $\widehat{\gamma}_j := \widehat{\gamma}_j(k_j)$, $\widehat{Q}_j := X_{n_j-k_j:n_j,j}$. For weight vectors $\omega = (\omega_1,\dots,\omega_m)^\top$ with $\sum_j \omega_j = 1$ (negative entries allowed), the **weighted pooled Hill estimator** is $\widehat{\gamma}_n(\omega) = \sum_{j=1}^m \omega_j \widehat{\gamma}_j$ as in DPS, Section 2.2. All limits are as $n\to\infty$ and $\xrightarrow{P}$, $\xrightarrow{d}$ denote convergence in probability and in distribution.

2.2 Assumptions

(A1) (Model.) The $X_{i,j}$, $1\le i\le n_j$, $1\le j\le m$, are i.i.d. (within and across machines) with common continuous distribution function F whose tail quantile function satisfies the second-order condition $\mathcal{C}_2(\gamma,\rho,A)$:

$$\lim_{t\to\infty}\frac{1}{A(t)}\left[\frac{U(tx)}{U(t)} - x^\gamma\right] \;=\; x^\gamma\,\frac{x^\rho-1}{\rho} \qquad \text{for all } x>0,$$

where $\gamma\in(0,1)$, $\rho<0$, and A has constant sign, $A(t)\to 0$, $|A|$ regularly varying with index ρ. Moreover $\mathbb{E}|X_-|<\infty$, where $X_- = \min(X,0)$.

(A2) (Subsample balance and intermediate sequences.) $k_j = k_j(n)\to\infty$, $k_j/n_j \to 0$ for each j; there are constants $b_j, c_j \in (0,\infty)$ (with $b_1 = c_1 = 1$) such that

$$\frac{n_1}{n_j} \to b_j, \qquad \frac{k_1}{k_j} \to c_j;$$

with $k := \sum_{j=1}^m k_j$, one has $\sqrt{k}\, A(n/k) \to \lambda \in \mathbb{R}$.

(A3) (Extreme level.) $p = p(n) \to 0$ with

$$\frac{k}{np} \to \infty \qquad \text{and} \qquad \frac{\sqrt{k}}{\log(k/(np))} \to \infty.$$

(A4) (Tail magnitude.) $\dfrac{\sqrt{k}}{U(n/k)} \to \mu \in [0, \infty)$.

2.3 Comments on the assumptions

On (A1). The professor's prescription "assume $\gamma > 0$" must be sharpened for expectiles: the *existence* of ξ_τ requires a finite first moment, hence $\gamma < 1$ together with $\mathbb{E}|X_-| < \infty$; and the proportionality constant $\varphi(\gamma) = (\gamma^{-1} - 1)^{-\gamma}$ is only defined for $\gamma \in (0, 1)$. The restriction $\gamma \in (0, 1)$ is therefore *intrinsic to the expectile problem* and not an artifact of the method. The requirement $\rho < 0$ is the same one made by DPS for their extreme quantile results (their Corollary 9) and guarantees that the Weissman extrapolation bias is of the order $O(|A(n_j/k_j)|)$. Note that, remarkably, **no restriction to $\gamma < 1/2$ is needed**: the QB route works on all of $(0, 1)$, whereas a LAWS-based route would require $\gamma < 1/2$ for $\sqrt{k_j}$-asymptotics (cf. Theorem 3.1 of Davison, Padoan and Stupfler, 2023).

On (A2). This is exactly the fixed-m distributed framework of DPS, Section 3: subsample sizes and effective sample sizes are asymptotically proportional. The condition $\sqrt{k}A(n/k) \to \lambda$ is the classical bias condition for the *combined* effective sample size k; Lemma 1 below shows it pins down all the machine-level biases. Note $k/n = \sum_j (k_j/n_j)(n_j/n) \to 0$ automatically.

On (A3). The first part says $1 - p$ is an extreme level relative to each machine (p is eventually far smaller than every k_j/n_j, see Lemma 1(iv)); the second part is the standard condition ensuring that extrapolation does not destroy the convergence rate, identical to DPS, Corollary 9. The leading case $np = O(1)$ — the target is beyond all subsamples and beyond the combined sample — is allowed.

On (A4). This condition controls the *expectile–quantile gap*: the relation $\xi_\tau \approx \varphi(\gamma) q_\tau$ has a remainder of order $1/q_\tau$ (Proposition D below), which after multiplication by $\sqrt{k_j}$ must remain bounded. It is the exact analogue, for the distributed/extreme setting, of the condition $\sqrt{n(1-\tau_n)}/q_{\tau_n} \to \lambda_2 \in \mathbb{R}$ in Theorem 3.5 of Davison, Padoan and Stupfler (2023) and Theorem 2.4 of Padoan and Stupfler (2022). Interpretation: for a Pareto-type tail $U(t) \sim Ct^\gamma$, (A4) holds with $\mu = 0$ as soon as $k = o\big(n^{2\gamma/(1+2\gamma)}\big)$, a mild undersmoothing requirement; the larger γ, the weaker the restriction. As the proofs show, neither μ nor the gap remainder appears in the limiting distribution: the extrapolation factor $D_n \to \infty$ *washes out* every $O(1)$ bias contribution, leaving only the Hill-type bias. This is the structural reason why the entire DPS weighting theory transfers verbatim to expectiles.

3. Preliminaries

3.1 Deterministic consequences of (A1)–(A4)

Define

$$S_0 := \sum_{i=1}^m c_i^{-1}, \qquad \kappa_j := c_j S_0, \qquad \pi_j := \frac{c_j^{-1}}{S_0}, \qquad d_j := \frac{c_j}{b_j} \cdot \frac{S_0}{\sum_{i=1}^m b_i^{-1}}, \qquad S_\alpha := \sum_{j=1}^m \frac{d_j^\alpha}{c_j} \;\; (\alpha \in \mathbb{R}).$$

Note that $(\pi_j)_j$ is a probability vector and that $S_\alpha = S_0 \sum_j \pi_j d_j^\alpha$; the notation is consistent at $\alpha = 0$.

> **Lemma 1 (rates across machines).** *Under (A1)–(A4), for every $j \in \{1, \ldots, m\}$:*
>
> *(i) $k/k_j \to \kappa_j \in (0, \infty)$ and $n/n_j \to b_j \sum_i b_i^{-1} \in (0, \infty)$;*
>
> *(ii) $\dfrac{n_j/k_j}{n/k} \to d_j \in (0, \infty)$, and $\sum_{j=1}^m \pi_j d_j = 1$;*

> *(iii)* $\sqrt{k_j}\,A(n_j/k_j) \to \lambda_j := \kappa_j^{-1/2} d_j^{\rho}\lambda$, *and* $\sqrt{k}\,A(n_j/k_j) \to d_j^{\rho}\lambda$;
>
> *(iv)* $d_{n,j} \to \infty$, $D_{n,j}/D_n \to 1$ *and* $\sqrt{k_j}/D_{n,j} \to \infty$;
>
> *(v)* $\limsup_n \sqrt{k_j}/U(1/p) \le \lim_n \sqrt{k_j}/U(n_j/k_j) = \kappa_j^{-1/2} d_j^{-\gamma}\mu < \infty$;
>
> *(vi)* $\limsup_n \sqrt{k_j}\,|A(1/p)| < \infty$.

Proof. *(i)* $k/k_j = \sum_i k_i/k_j \to \sum_i c_j/c_i = c_j S_0 = \kappa_j$; similarly $n/n_j = \sum_i n_i/n_j \to b_j \sum_i b_i^{-1}$.

(ii) Write $\dfrac{n_j/k_j}{n/k} = \dfrac{n_j}{n}\cdot\dfrac{k}{k_j} \to \dfrac{1}{b_j\sum_i b_i^{-1}}\cdot c_j S_0 = d_j$. Moreover
$\sum_j \pi_j d_j = \dfrac{1}{S_0}\sum_j \dfrac{d_j}{c_j} = \dfrac{1}{\sum_i b_i^{-1}}\sum_j b_j^{-1} = 1.$

(iii) Since $|A|$ is regularly varying with index ρ and A has constant sign, the locally uniform convergence theorem for regularly varying functions (Bingham, Goldie and Teugels, 1987, Theorem 1.5.2; de Haan and Ferreira, 2006, Proposition B.1.4) yields $A(ts_t)/A(t) \to s^{\rho}$ whenever $s_t \to s \in (0,\infty)$. Taking $t = n/k$ and $s_t = (n_j/k_j)/(n/k) \to d_j$ by (ii) gives $A(n_j/k_j)/A(n/k) \to d_j^{\rho}$, whence by (A2)

$$\sqrt{k_j}\,A(n_j/k_j) = \sqrt{\tfrac{k_j}{k}}\cdot\sqrt{k}A(n/k)\cdot\frac{A(n_j/k_j)}{A(n/k)} \longrightarrow \kappa_j^{-1/2}\,\lambda\, d_j^{\rho},$$

and similarly $\sqrt{k}A(n_j/k_j) \to d_j^{\rho}\lambda$.

(iv) $d_{n,j} = d_n\cdot\dfrac{k_j\,n}{k\,n_j}$ and $\dfrac{k_j n}{k n_j} \to d_j^{-1} \in (0,\infty)$ by (i)–(ii), so $d_{n,j} \to \infty$ by (A3); moreover $D_{n,j} - D_n = \log\dfrac{k_j n}{k n_j} \to -\log d_j$ is bounded while $D_n \to \infty$, so $D_{n,j}/D_n \to 1$. Finally $\sqrt{k_j}/D_{n,j} = \sqrt{k_j/k}\,(D_n/D_{n,j})\,\sqrt{k}/D_n \to \infty$ by (A3).

(v) For n large, $1/p \ge n_j/k_j$ by (iv), and U is nondecreasing, so $U(1/p) \ge U(n_j/k_j)$ eventually, whence the inequality. For the limit: U is regularly varying with index γ (a consequence of (A1)), so by the locally uniform convergence theorem again, $U(n_j/k_j)/U(n/k) \to d_j^{\gamma}$; then $\sqrt{k_j}/U(n_j/k_j) = \sqrt{k_j/k}\cdot\big(\sqrt{k}/U(n/k)\big)\cdot U(n/k)/U(n_j/k_j) \to \kappa_j^{-1/2}\mu\, d_j^{-\gamma}$ by (A4).

(vi) Since $|A|$ is regularly varying with index $\rho < 0$, the Potter bounds (Bingham, Goldie and Teugels, 1987, Theorem 1.5.6; de Haan and Ferreira, 2006, Proposition B.1.9.5) give a t_0 with

$$\frac{|A(tx)|}{|A(t)|} \le 2\,x^{\rho/2} \le 2 \qquad \text{for all } t \ge t_0,\ x \ge 1.$$

Apply this with $t = n_j/k_j \to \infty$ and $tx = 1/p$ (legitimate eventually, since $1/p \ge n_j/k_j$ by (iv)): $|A(1/p)| \le 2|A(n_j/k_j)|$ eventually, so $\limsup_n \sqrt{k_j}|A(1/p)| \le 2|\lambda_j| < \infty$ by (iii). ■

3.2 Imported results

The following are quoted with precise references; they are the only external ingredients.

> **Proposition A (Hill estimator; de Haan and Ferreira, 2006, Theorem 3.2.5).** *Let $Y_1,\dots,Y_N$ be i.i.d. with tail quantile function satisfying $\mathcal{C}_2(\gamma,\rho,A)$, $\gamma > 0$. If $K \to \infty$, $K/N \to 0$ and $\sqrt{K}A(N/K) \to \lambda' \in \mathbb{R}$, then the Hill estimator $\widehat{\gamma}(K)$ built on the top K order statistics satisfies*
>
> $$\sqrt{K}\big(\widehat{\gamma}(K) - \gamma\big) \xrightarrow{d} \mathcal{N}\Big(\frac{\lambda'}{1-\rho},\, \gamma^2\Big).$$

> **Proposition B (intermediate order statistic; de Haan and Ferreira, 2006, Theorem 2.4.1 taken at $s = 1$).** *In the setting of Proposition A,*
>
> $$\sqrt{K}\Big(\frac{Y_{N-K:N}}{U(N/K)} - 1\Big) \xrightarrow{d} \mathcal{N}(0,\gamma^2); \qquad \text{in particular} \qquad \frac{Y_{N-K:N}}{U(N/K)} - 1 = O_P\big(K^{-1/2}\big).$$

> **Proposition C (uniform second-order bound for $\log U$; de Haan and Ferreira, 2006, Theorem B.3.10 in its $\log U$ form, as used in the proof of their Theorem 4.3.8).** *Under $\mathcal{C}_2(\gamma,\rho,A)$ with $\rho < 0$: for every $\varepsilon > 0$ there is $t_0 = t_0(\varepsilon)$ such that for all $t \ge t_0$ and $x \ge 1$,*

$$\left| \frac{\log U(tx) - \log U(t) - \gamma \log x}{A(t)} - \frac{x^\rho - 1}{\rho} \right| \ \le\ \varepsilon\, x^{\rho+\varepsilon}.$$

Proposition D (expectile–quantile gap; Daouia, Girard and Stupfler, 2020, Proposition 1(i); see also Corollary 3.4 of Davison, Padoan and Stupfler, 2023, where exactly this form is used). *Under (A1), as $\tau \uparrow 1$,*

$$\frac{\xi_\tau}{q_\tau} \ =\ \varphi(\gamma)\,\big(1 + r(\tau)\big), \qquad r(\tau) = \gamma(\gamma^{-1}-1)^\gamma\, \frac{\mathbb{E}(X)}{q_\tau}\big(1+o(1)\big) + \left[\frac{(\gamma^{-1}-1)^{-\rho}}{1-\gamma-\rho} + \frac{(\gamma^{-1}-1)^{-\rho}-1}{\rho}\right] A\big((1-\tau)^{-1}\big)\big(1+o(1)\big).$$

In particular $r(\tau) = O\big(1/q_\tau\big) + O\big(|A((1-\tau)^{-1})|\big) \to 0$. (Only this order statement is used in the proofs below.)

Proposition E (oracle property of the variance-optimal pooled Hill estimator; DPS, Theorem 3). *Under (A1)–(A2), let $\widehat{\gamma}_n^{\text{or}} = \widehat{\gamma}_n^{\text{or}}(k)$ denote the (unfeasible) Hill estimator computed from the top $k = \sum_j k_j$ order statistics of the combined n-sample, and let $\widetilde{\omega}_n^{(\text{Var})} = (k_1/k, \ldots, k_m/k)^\top$. If*

$$\frac{k_j}{n_j} = \frac{k}{n}\Big(1 + O\big(k^{-1/2}\big)\Big) \quad \text{for all } j,$$

then $\sqrt{k}\big(\widehat{\gamma}_n(\widetilde{\omega}_n^{(\text{Var})}) - \widehat{\gamma}_n^{\text{or}}\big) = o_P(1)$.

Remark 1 (well-definedness). Heavy tails ($\gamma > 0$) imply $U(t) \to \infty$, so $\widehat{Q}_j \xrightarrow{P} \infty$ and in particular $\widehat{Q}_j > 0$ with probability tending to one: logarithms of the transmitted statistics are well defined eventually. Likewise, since $\widehat{\gamma}_j \xrightarrow{P} \gamma \in (0,1)$ (Proposition A), the event $\{\widehat{\gamma}_j \in (0,1)\ \forall j\}$ has probability tending to one; on its complement the estimators below may be defined arbitrarily (say, by truncating $\widehat{\gamma}_j$ to $[\epsilon, 1-\epsilon]$). All asymptotic statements are unaffected. These conventions are used without further comment.

4. Construction of the pooled extreme expectile estimator

4.1 Machine-level estimators from the transmitted statistics

The transmitted pair $(\widehat{\gamma}_j, \widehat{Q}_j)$ determines the Weissman (1978) extreme *quantile* estimator of machine j,

$$\hat{q}_j^*(1-p) \ =\ \left(\frac{k_j}{n_j p}\right)^{\widehat{\gamma}_j} \widehat{Q}_j \ =\ d_{n,j}^{\widehat{\gamma}_j}\, X_{n_j-k_j:n_j,j},$$

exactly as in DPS. The Bellini proportionality $\xi_\tau \sim \varphi(\gamma)\, q_\tau$ ($\tau \uparrow 1$; Proposition D) suggests converting it into the **machine-level QB extrapolating expectile estimator**

$$\boxed{\hat{\xi}_j^*(1-p) \ =\ \varphi(\widehat{\gamma}_j)\, \left(\frac{k_j}{n_j p}\right)^{\widehat{\gamma}_j} X_{n_j-k_j:n_j,j} \ =\ \left(\widehat{\gamma}_j^{-1} - 1\right)^{-\widehat{\gamma}_j}\, \hat{q}_j^*(1-p).}$$

This is the one-sample estimator of Daouia, Girard and Stupfler (2018, 2020), studied at extreme levels in Davison, Padoan and Stupfler (2023, Section 3.3) and Padoan and Stupfler (2022, Section 2.3) — computed here *exclusively from the transmitted statistics T_j*.

4.2 Geometric pooling

For a weight vector ω with $\sum_j \omega_j = 1$, the **pooled extreme expectile estimator** is the weighted *geometric* mean

$$\boxed{\hat{\xi}_n^*(1-p\,|\,\omega) \ =\ \prod_{j=1}^m \left[\hat{\xi}_j^*(1-p)\right]^{\omega_j}.}$$

Why geometric and not arithmetic? Both the Weissman factor $d_{n,j}^{\widehat{\gamma}_j}$ and the Bellini factor $\varphi(\widehat{\gamma}_j)$ are *exponential* in $\widehat{\gamma}_j$; on the $\log$ scale the estimator is exactly *linear* in the transmitted Hill estimators:

$$\log \hat{\xi}_n^*(1-p\,|\,\omega) = \sum_{j=1}^m \omega_j \Big[\log \varphi(\widehat{\gamma}_j) + \widehat{\gamma}_j\, D_{n,j} + \log \widehat{Q}_j\Big].$$

The dominating term is $\sum_j \omega_j \widehat{\gamma}_j D_{n,j} \approx \widehat{\gamma}_n(\omega)\, D_n$: geometric pooling puts the problem back on the *standard scale of the pooled Hill estimator* $\widehat{\gamma}_n(\omega)$, which is the precise mechanism exploited by DPS (Section 2.3) for quantiles; an arithmetic average would break this exact linearization (see Remark 5 in Section 8 below). The log scale is also the natural scale for inference about extreme tail quantities (Drees, 2003).

Theorem 2 below makes the heuristic rigorous: the $\sqrt{k}/D_n$-asymptotics of $\hat{\xi}_n^*(1-p\,|\,\omega)$ coincide with those of $\widehat{\gamma}_n(\omega)$, so weight optimization for the *expectile* problem reduces *exactly* to weight optimization for the pooled Hill estimator.

5. Asymptotic theory

5.1 Machine-level (joint) asymptotics

> **Theorem 1 (marginal QB extreme expectile estimators, jointly across machines).** *Assume (A1)–(A4). Then, with $\lambda_j = \kappa_j^{-1/2} d_j^{\rho} \lambda$ as in Lemma 1(iii),*
>
> $$\left(\frac{\sqrt{k_j}}{D_{n,j}}\Big(\frac{\hat{\xi}_j^*(1-p)}{\xi_{1-p}} - 1\Big)\right)_{1\le j\le m} = \Big(\sqrt{k_j}\,(\widehat{\gamma}_j - \gamma)\Big)_{1\le j\le m} + o_P(1) \xrightarrow{d} \mathcal{N}_m\bigg(\Big(\frac{\lambda_j}{1-\rho}\Big)_{1\le j\le m},\ \gamma^2 I_m\bigg).$$

Proof. Fix j and abbreviate $t_j := n_j/k_j$, so that $1/p = t_j\, d_{n,j}$ and $q_{1-p} = U(1/p)$. On the event of Remark 1 (probability $\to 1$) we have the *exact* decomposition

$$\log \hat{\xi}_j^*(1-p) - \log \xi_{1-p} = T_1 + T_2 + T_3 + T_4 + T_5,$$

with

$$T_1 = \log\varphi(\widehat{\gamma}_j) - \log\varphi(\gamma), \qquad T_2 = (\widehat{\gamma}_j - \gamma)\, D_{n,j}, \qquad T_3 = \log\frac{\widehat{Q}_j}{U(t_j)},$$
$$T_4 = -\Big[\log U(t_j d_{n,j}) - \log U(t_j) - \gamma D_{n,j}\Big], \quad T_5 = \log\big(\varphi(\gamma)\, U(1/p)\big) - \log \xi_{1-p},$$

as is checked by direct cancellation (the terms $\pm\log U(t_j)$, $\pm\gamma D_{n,j}$, $\pm\log\varphi(\gamma)$ and $\pm\log U(1/p)$ all cancel, leaving $\log\varphi(\widehat{\gamma}_j) + \widehat{\gamma}_j D_{n,j} + \log\widehat{Q}_j - \log\xi_{1-p}$).

Term T_1. By Proposition A (applicable on machine j with $N = n_j$, $K = k_j$, $\lambda' = \lambda_j$, thanks to Lemma 1(iii)), $\widehat{\gamma}_j - \gamma = O_P(k_j^{-1/2})$; in particular $\widehat{\gamma}_j \xrightarrow{P} \gamma$. Since $\log\varphi$ is continuously differentiable on $(0,1)$ with derivative m, the mean value theorem gives, on an event of probability tending to one, $T_1 = m(\bar{\gamma}_j)(\widehat{\gamma}_j - \gamma)$ with $\bar{\gamma}_j$ between $\widehat{\gamma}_j$ and γ, and $m(\bar{\gamma}_j) \xrightarrow{P} m(\gamma)$ by continuity. Hence

$$T_1 = O_P\big(k_j^{-1/2}\big).$$

Term T_3. By Proposition B on machine j, $\widehat{Q}_j/U(t_j) - 1 = O_P(k_j^{-1/2}) = o_P(1)$, so $T_3 = \log\big(1 + O_P(k_j^{-1/2})\big) = O_P\big(k_j^{-1/2}\big)$.

Term T_4. Apply Proposition C with $\varepsilon \in (0, -\rho)$, $t = t_j \to \infty$ and $x = d_{n,j} \ge 1$ (eventually, by Lemma 1(iv)):

$$T_4 = -A(t_j)\left[\frac{d_{n,j}^{\rho} - 1}{\rho} + O\big(\varepsilon\, d_{n,j}^{\rho+\varepsilon}\big)\right] = -A(t_j)\Big[-\frac{1}{\rho} + o(1)\Big] = O\big(|A(t_j)|\big) = O\big(k_j^{-1/2}\big),$$

the last two steps because $d_{n,j} \to \infty$ with $\rho + \varepsilon < 0$, and because $\sqrt{k_j} A(t_j) \to \lambda_j$ (Lemma 1(iii)).

Term T_5. By Proposition D applied at $\tau = 1 - p \uparrow 1$, $T_5 = -\log\big(1 + r(1-p)\big)$ with $r(1-p) \to 0$, so $|T_5| \le 2|r(1-p)|$ eventually, and

$$\sqrt{k_j}\,|T_5| \ \le\ 2\sqrt{k_j}\,\Big[O\big(1/U(1/p)\big) + O\big(|A(1/p)|\big)\Big] \ =\ O(1)$$

by Lemma 1(v)–(vi). Thus $T_5 = O(k_j^{-1/2})$.

Collecting. $\sqrt{k_j}\big[\log\hat{\xi}_j^*(1-p) - \log\xi_{1-p}\big] = \sqrt{k_j}(\widehat{\gamma}_j - \gamma)\, D_{n,j} + O_P(1)$, and dividing by $D_{n,j} \to \infty$,

$$\frac{\sqrt{k_j}}{D_{n,j}}\log\frac{\hat{\xi}_j^*(1-p)}{\xi_{1-p}} = \sqrt{k_j}(\widehat{\gamma}_j - \gamma) + o_P(1). \tag{5.1}$$

The right-hand side is $O_P(1)$, and $D_{n,j}/\sqrt{k_j} \to 0$ by Lemma 1(iv); hence $\log\big(\hat{\xi}_j^*/\xi_{1-p}\big) = o_P(1)$, i.e. $\hat{\xi}_j^*/\xi_{1-p} \xrightarrow{P} 1$, and by $e^u - 1 = u(1+o(1))$ as $u \to 0$, the relative error inherits the expansion (5.1):

$$\frac{\sqrt{k_j}}{D_{n,j}}\Big(\frac{\hat{\xi}_j^*(1-p)}{\xi_{1-p}} - 1\Big) = \sqrt{k_j}(\widehat{\gamma}_j - \gamma) + o_P(1).$$

Since m is fixed, the vector statement holds with a vector $o_P(1)$. Finally, the data are independent across machines, so the statistics $(\widehat{\gamma}_j)_{1\le j\le m}$ are mutually independent; Proposition A on each machine and independence give the joint limit $\mathcal{N}_m\big((\lambda_j/(1-\rho))_j, \gamma^2 I_m\big)$. ■

5.2 Pooled asymptotics with deterministic or random weights

> **Theorem 2 (pooled QB extreme expectile estimator).** *Assume (A1)–(A4). Let $\omega \in \mathbb{R}^m$ with $\sum_j \omega_j = 1$ and let $\widehat{\omega}_n$ be a (possibly data-dependent) weight vector with $\sum_j \widehat{\omega}_{n,j} = 1$ almost surely and $\widehat{\omega}_n \xrightarrow{P} \omega$. Then*
>
> $$\frac{\sqrt{k}}{D_n}\Big(\frac{\hat{\xi}_n^*(1-p\,|\,\widehat{\omega}_n)}{\xi_{1-p}} - 1\Big) = \sqrt{k}\,\big(\widehat{\gamma}_n(\omega) - \gamma\big) + o_P(1) \;\xrightarrow{d}\; \mathcal{N}\left(\frac{\lambda}{1-\rho}\sum_{j=1}^m \omega_j d_j^\rho,\;\; \gamma^2\, S_0 \sum_{j=1}^m c_j\omega_j^2\right).$$

Proof. *Step 1 (deterministic weights).* Using $\sum_j \omega_j = 1$,

$$\frac{\sqrt{k}}{D_n}\log\frac{\hat{\xi}_n^*(1-p\,|\,\omega)}{\xi_{1-p}} = \sum_{j=1}^m \omega_j\,\frac{\sqrt{k}}{\sqrt{k_j}}\cdot\frac{D_{n,j}}{D_n}\cdot\frac{\sqrt{k_j}}{D_{n,j}}\log\frac{\hat{\xi}_j^*(1-p)}{\xi_{1-p}}.$$

By (5.1), $\dfrac{\sqrt{k_j}}{D_{n,j}}\log\dfrac{\hat{\xi}_j^*}{\xi_{1-p}} = \sqrt{k_j}(\widehat{\gamma}_j - \gamma) + o_P(1)$; by Lemma 1, $\sqrt{k/k_j} \to \kappa_j^{1/2} < \infty$ and $D_{n,j}/D_n \to 1$, so with $\varepsilon_{n,j} := \sqrt{k/k_j}\,(D_{n,j}/D_n) - \sqrt{k/k_j} \to 0$ deterministic and $\sqrt{k}(\widehat{\gamma}_j - \gamma) = \sqrt{k/k_j}\,\sqrt{k_j}(\widehat{\gamma}_j - \gamma) = O_P(1)$:

$$\frac{\sqrt{k}}{D_n}\log\frac{\hat{\xi}_n^*(1-p\,|\,\omega)}{\xi_{1-p}} = \sum_{j=1}^m \omega_j\,\sqrt{k}\,(\widehat{\gamma}_j - \gamma) + o_P(1) = \sqrt{k}\,\big(\widehat{\gamma}_n(\omega) - \gamma\big) + o_P(1). \tag{5.2}$$

Step 2 (random weights). Since $\sum_j(\widehat{\omega}_{n,j} - \omega_j) = 0$,

$$\log\hat{\xi}_n^*(1-p\,|\,\widehat{\omega}_n) - \log\hat{\xi}_n^*(1-p\,|\,\omega) = \sum_{j=1}^m(\widehat{\omega}_{n,j} - \omega_j)\Big[\log\hat{\xi}_j^*(1-p) - \log\xi_{1-p}\Big].$$

Each bracket is $O_P(D_{n,j}/\sqrt{k_j}) = O_P(D_n/\sqrt{k})$ by (5.1) and Lemma 1, and $\widehat{\omega}_{n,j} - \omega_j = o_P(1)$; multiplying by $\sqrt{k}/D_n$ leaves $\sum_j o_P(1)O_P(1) = o_P(1)$. Combining with (5.2),

$$\frac{\sqrt{k}}{D_n}\log\frac{\hat{\xi}_n^*(1-p\,|\,\widehat{\omega}_n)}{\xi_{1-p}} = \sqrt{k}\,\big(\widehat{\gamma}_n(\omega) - \gamma\big) + o_P(1).$$

As in the proof of Theorem 1, the left-hand side being $O_P(1)$ and $D_n/\sqrt{k} \to 0$ imply $\hat{\xi}_n^*(1-p\,|\,\widehat{\omega}_n)/\xi_{1-p} \xrightarrow{P} 1$, and the same expansion holds for the relative error.

Step 3 (limit distribution of the pooled Hill estimator). Write

$$\sqrt{k}\,\big(\widehat{\gamma}_n(\omega)-\gamma\big)=\sum_{j=1}^m \omega_j\sqrt{\frac{k}{k_j}}\;\sqrt{k_j}\big(\widehat{\gamma}_j-\gamma\big).$$

The m summands are independent (independence across machines), $\sqrt{k/k_j}\to\kappa_j^{1/2}$, and $\sqrt{k_j}(\widehat{\gamma}_j-\gamma)\xrightarrow{d}\mathcal{N}(\lambda_j/(1-\rho),\gamma^2)$ by Proposition A and Lemma 1(iii). Hence

$$\sqrt{k}\,\big(\widehat{\gamma}_n(\omega)-\gamma\big)\xrightarrow{d}\mathcal{N}\left(\sum_j \omega_j\,\kappa_j^{1/2}\,\frac{\lambda_j}{1-\rho},\ \gamma^2\sum_j \omega_j^2\,\kappa_j\right),$$

and the parameters simplify via $\kappa_j^{1/2}\lambda_j=d_j^\rho\lambda$ and $\kappa_j=c_jS_0$:

$$\text{bias}=\frac{\lambda}{1-\rho}\sum_j \omega_j d_j^\rho,\qquad \text{variance}=\gamma^2\,S_0\sum_j c_j\omega_j^2.\qquad\blacksquare$$

Remark 2 (reduction principle). Theorem 2 is the central structural fact: *to first order, estimating the extreme expectile in the distributed setting is exactly the problem of estimating γ by weighted pooling.* The order statistics $\widehat{Q}_j$, the Bellini factors, the Weissman extrapolation bias and the expectile–quantile gap all contribute $O_P(\sqrt{k_j}^{-1})$ terms that the normalization $\sqrt{k}/D_n$ annihilates. Consequently the optimal-weight theory of DPS applies *verbatim*, as developed next — this is the expectile counterpart of their Corollary 9 for extreme quantiles.

6. Optimal pooling

By Theorem 2, the asymptotic bias and variance of $\dfrac{\sqrt{k}}{D_n}\Big(\dfrac{\hat{\xi}_n^*(1-p|\omega)}{\xi_{1-p}}-1\Big)$ are

$$\omega^\top B_c\qquad\text{and}\qquad \omega^\top V_c\,\omega,\qquad \text{where}\quad B_c:=\beta\,\big(d_1^\rho,\ldots,d_m^\rho\big)^\top,\ \ \beta:=\frac{\lambda}{1-\rho},\ \ V_c:=\gamma^2 S_0\,\mathrm{diag}(c_1,\ldots,c_m),$$

the same matrices as in DPS, Section 3. We optimize over $\mathcal{W}:=\{\omega\in\mathbb{R}^m:\ \mathbf{1}^\top\omega=1\}$.

6.1 Variance-optimal pooling

> **Proposition 3 (variance-optimal weights).** *The unique minimizer of $\omega\mapsto\omega^\top V_c\,\omega$ over $\mathcal{W}$ is*
>
> $$\omega_j^{(\mathrm{Var})}=\frac{c_j^{-1}}{S_0}=\pi_j,\qquad \text{with minimal variance}\qquad \big(\omega^{(\mathrm{Var})}\big)^\top V_c\,\omega^{(\mathrm{Var})}=\gamma^2.$$
>
> *Moreover, the data-driven weights $\widetilde{\omega}_n^{(\mathrm{Var})}:=(k_1/k,\ldots,k_m/k)^\top$ — computable from the transmitted T_j alone — satisfy $\widetilde{\omega}_n^{(\mathrm{Var})}\to\omega^{(\mathrm{Var})}$ (deterministically).*

Proof. By the Cauchy–Schwarz inequality, for any $\omega\in\mathcal{W}$,

$$1=\Big(\sum_j\omega_j\Big)^2=\Big(\sum_j\big(\sqrt{c_j}\,\omega_j\big)\,c_j^{-1/2}\Big)^2\le\Big(\sum_j c_j\omega_j^2\Big)\Big(\sum_j c_j^{-1}\Big)=S_0\sum_j c_j\omega_j^2,$$

i.e. $\omega^\top V_c\,\omega=\gamma^2 S_0\sum_j c_j\omega_j^2\ge\gamma^2$, with equality if and only if $\sqrt{c_j}\,\omega_j\propto c_j^{-1/2}$, i.e. $\omega_j\propto c_j^{-1}$; the constraint then forces $\omega_j=c_j^{-1}/S_0$. The last claim follows from $k_j/k\to\kappa_j^{-1}=(c_jS_0)^{-1}$ (Lemma 1(i)). $\blacksquare$

> **Corollary 4 (variance-optimal distributed extreme expectile estimator).** *Under (A1)–(A4),*
>
> $$\frac{\sqrt{k}}{D_n}\Big(\frac{\hat{\xi}_n^*\big(1-p\,|\,\widetilde{\omega}_n^{(\mathrm{Var})}\big)}{\xi_{1-p}}-1\Big)\ \xrightarrow{d}\ \mathcal{N}\left(\frac{\lambda}{1-\rho}\cdot\frac{S_\rho}{S_0},\ \gamma^2\right).$$

Proof. Theorem 2 with $\widehat{\omega}_n = \widetilde{\omega}_n^{(\mathrm{Var})} \to \omega^{(\mathrm{Var})} = (\pi_j)_j$: the bias is $\beta \sum_j \pi_j d_j^\rho = \beta S_\rho / S_0$ and the variance is γ^2 by Proposition 3. ■

6.2 The infeasible benchmark and the oracle property

The natural benchmark is the QB extreme expectile estimator one *would* compute if the n observations could be combined, with effective sample size $k = \sum_j k_j$:

$$\hat{\xi}_n^{*,\mathrm{or}}(1-p) := \varphi\big(\widehat{\gamma}_n^{\mathrm{or}}\big)\Big(\frac{k}{np}\Big)^{\widehat{\gamma}_n^{\mathrm{or}}} X_{n-k:n},$$

where $\widehat{\gamma}_n^{\mathrm{or}} = \widehat{\gamma}_n^{\mathrm{or}}(k)$ is the Hill estimator of the combined sample and $X_{n-k:n}$ its $(n-k)$-th order statistic.

> **Theorem 5 (benchmark and oracle property).** *Assume (A1)–(A4).*
>
> *(i) (Benchmark CLT.)* $\dfrac{\sqrt{k}}{D_n}\Big(\dfrac{\hat{\xi}_n^{*,\mathrm{or}}(1-p)}{\xi_{1-p}} - 1\Big) = \sqrt{k}\big(\widehat{\gamma}_n^{\mathrm{or}} - \gamma\big) + o_P(1) \xrightarrow{d} \mathcal{N}\Big(\dfrac{\lambda}{1-\rho}, \gamma^2\Big).$
>
> *(ii) (Bias comparison.) The variance-optimal distributed estimator of Corollary 4 attains the benchmark variance γ^2, and its absolute asymptotic bias dominates that of the benchmark:*
>
> $$\Big|\frac{\lambda}{1-\rho}\,\frac{S_\rho}{S_0}\Big| \;\ge\; \Big|\frac{\lambda}{1-\rho}\Big|,$$
>
> *with equality if and only if $\lambda = 0$ or $d_j = 1$ for all j (asymptotically equal sample fractions k_j/n_j).*
>
> *(iii) (Full asymptotic equivalence.) If in addition $k_j/n_j = (k/n)\big(1 + O(k^{-1/2})\big)$ for all j, then*
>
> $$\frac{\sqrt{k}}{D_n}\left(\frac{\hat{\xi}_n^{*}\big(1-p\,|\,\widetilde{\omega}_n^{(\mathrm{Var})}\big)}{\hat{\xi}_n^{*,\mathrm{or}}(1-p)} - 1\right) = o_P(1),$$
>
> *i.e. the feasible distributed estimator is asymptotically indistinguishable, at the rate, from the infeasible benchmark.*

Proof. *(i)* Apply Theorem 1 with $m = 1$ "machine" equal to the combined sample: the required conditions are $k \to \infty$, $k/n \to 0$ (Lemma 1 / (A2)), $\sqrt{k}A(n/k) \to \lambda$ (A2), $k/(np) \to \infty$ and $\sqrt{k}/D_n \to \infty$ (A3), and $\sqrt{k}/U(n/k) \to \mu$ (A4) — all hold, with $d_1 = c_1 = 1$, $\kappa_1 = 1$, $\lambda_1 = \lambda$. Proposition A gives the limit.

(ii) The variance statement is Proposition 3. For the bias: since $x \mapsto x^\rho$ is convex on $(0,\infty)$ (as $\rho < 0$) and $(\pi_j)_j$ is a probability vector with $\sum_j \pi_j d_j = 1$ (Lemma 1(ii)), Jensen's inequality gives

$$\frac{S_\rho}{S_0} = \sum_j \pi_j\, d_j^\rho \;\ge\; \Big(\sum_j \pi_j d_j\Big)^\rho = 1,$$

with equality if and only if the d_j are all equal (strict convexity), which combined with $\sum_j \pi_j d_j = 1$ forces $d_j \equiv 1$. Multiplying both sides of $S_\rho/S_0 \ge 1$ by $|\beta| = |\lambda|/(1-\rho)$ proves the claim, with equality iff $\lambda = 0$ or $d_j \equiv 1$. Finally, $d_j \equiv 1$ is exactly asymptotic equality of the sample fractions, since $d_j = \lim\,(k/n)/(k_j/n_j)$.

(iii) The premise implies $(k/n)/(k_j/n_j) \to 1$, i.e. $d_j = 1$ for all j, so $\omega^{(\mathrm{Var})} = (\pi_j)_j$ and all conditions used so far stand. By (5.2)–Step 2 of the proof of Theorem 2 (with $\widehat{\omega}_n = \widetilde{\omega}_n^{(\mathrm{Var})} \to \omega^{(\mathrm{Var})}$) and by part (i),

$$\frac{\sqrt{k}}{D_n}\log\frac{\hat{\xi}_n^{*}\big(1-p\,|\,\widetilde{\omega}_n^{(\mathrm{Var})}\big)}{\xi_{1-p}} = \sqrt{k}\big(\widehat{\gamma}_n(\widetilde{\omega}_n^{(\mathrm{Var})}) - \gamma\big) + o_P(1), \qquad \frac{\sqrt{k}}{D_n}\log\frac{\hat{\xi}_n^{*,\mathrm{or}}(1-p)}{\xi_{1-p}} = \sqrt{k}\big(\widehat{\gamma}_n^{\mathrm{or}} - \gamma\big) + o_P(1),$$

where in the first display we replaced $\widehat{\gamma}_n(\omega^{(\mathrm{Var})})$ (delivered by Theorem 2's expansion) with $\widehat{\gamma}_n(\widetilde{\omega}_n^{(\mathrm{Var})})$, which is legitimate at this order since $\sqrt{k}\big(\widehat{\gamma}_n(\widetilde{\omega}_n^{(\mathrm{Var})}) - \widehat{\gamma}_n(\omega^{(\mathrm{Var})})\big) = \sum_j (\widetilde{\omega}_{n,j} - \pi_j)\,\sqrt{k}(\widehat{\gamma}_j - \gamma) = o(1)\,O_P(1) = o_P(1)$ (deterministic weights $\widetilde{\omega}_{n,j} \to \pi_j$). Subtracting and invoking Proposition E,

$$\frac{\sqrt{k}}{D_n}\log\frac{\hat{\xi}_n^*(1-p\,|\,\widetilde{\omega}_n^{(\mathrm{Var})})}{\hat{\xi}_n^{*,\mathrm{or}}(1-p)}=\sqrt{k}\big(\widehat{\gamma}_n(\widetilde{\omega}_n^{(\mathrm{Var})})-\widehat{\gamma}_n^{\mathrm{or}}\big)+o_P(1)=o_P(1).$$

Both estimators are consistent for ξ_{1-p} in relative terms (Theorem 2 and part (i)), so their ratio tends to 1 in probability, and the $\log\mapsto$ relative-error conversion of the proof of Theorem 1 applies once more. ■

6.3 AMSE-optimal pooling

Define the **asymptotic mean squared error coefficient** of the pooled estimator with weights $\omega\in\mathcal{W}$ as the (squared bias + variance) of the limit law in Theorem 2:

$$M(\omega):=\big(\omega^\top B_c\big)^2+\omega^\top V_c\,\omega,\qquad\text{so that}\qquad\mathrm{AMSE}\big(\hat{\xi}_n^*(1-p|\omega)\big)=\frac{D_n^2}{k}\,M(\omega)\,\big(1+o(1)\big)$$

in the usual asymptotic sense; the factor D_n^2/k is common to all weight choices, so weights are compared through $M(\cdot)$. The benchmark coefficient is $M^{\mathrm{or}}:=\beta^2+\gamma^2$ by Theorem 5(i).

> **Theorem 6 (AMSE-optimal weights, value, and comparison with the benchmark).** *Assume (A1)–(A4), and recall $\beta=\lambda/(1-\rho)$, $S_\alpha=\sum_j d_j^\alpha/c_j$.*
>
> *(i) The map M has a unique minimizer over $\mathcal{W}$, namely*
>
> $$\omega^{(\mathrm{AMSE})}=\frac{\big(1+B_c^\top V_c^{-1}B_c\big)V_c^{-1}\mathbf{1}-\big(\mathbf{1}^\top V_c^{-1}B_c\big)V_c^{-1}B_c}{\big(1+B_c^\top V_c^{-1}B_c\big)\big(\mathbf{1}^\top V_c^{-1}\mathbf{1}\big)-\big(\mathbf{1}^\top V_c^{-1}B_c\big)^2},\qquad\text{explicitly}\qquad\omega_j^{(\mathrm{AMSE})}=\frac{c_j^{-1}\Big[\gamma^2S_0+\beta^2S_{2\rho}-\beta^2S_\rho\,d_j^\rho\Big]}{\Delta},$$
>
> *where $\Delta:=\big(\gamma^2S_0+\beta^2S_{2\rho}\big)S_0-\beta^2S_\rho^2>0$, with optimal value*
>
> $$M^*:=M\big(\omega^{(\mathrm{AMSE})}\big)=\frac{\gamma^2S_0\big(\gamma^2S_0+\beta^2S_{2\rho}\big)}{\Delta}.$$
>
> *(ii) Suppose the d_j are not all equal to 1 (unbalanced design). Then*
>
> $$M^*\;\geq\;M^{\mathrm{or}}=\beta^2+\gamma^2\qquad\Longleftrightarrow\qquad|\lambda|\;\leq\;\lambda_0:=\gamma(1-\rho)\sqrt{\frac{S_\rho^2-S_0^2}{S_0S_{2\rho}-S_\rho^2}}\;\in(0,\infty).$$
>
> *In particular, the AMSE-optimal distributed extreme expectile estimator strictly beats the infeasible benchmark in AMSE whenever the bias is large, $|\lambda|>\lambda_0$ — the exact expectile analogue of DPS, Theorem 4.*

Proof. *(i) Existence, uniqueness, first-order conditions.* $M(\omega)=\omega^\top\big(V_c+B_cB_c^\top\big)\omega$ with $V_c\succ 0$ (as $\gamma,c_j,S_0>0$) and $B_cB_c^\top\succeq 0$, so M is strictly convex on $\mathbb{R}^m$ and has a unique minimizer on the affine set $\mathcal{W}$, characterized by the Lagrange condition

$$2\big(\omega^\top B_c\big)B_c+2V_c\,\omega=\nu\,\mathbf{1},\qquad\mathbf{1}^\top\omega=1,$$

for some $\nu\in\mathbb{R}$. Set $a:=\omega^\top B_c$. Solving the stationarity equation for ω:

$$\omega=\frac{\nu}{2}V_c^{-1}\mathbf{1}-a\,V_c^{-1}B_c.\tag{6.1}$$

Multiplying (6.1) by $B_c^\top$ gives $a=\frac{\nu}{2}\,\mathbf{1}^\top V_c^{-1}B_c-a\,B_c^\top V_c^{-1}B_c$, i.e.

$$a=\frac{\nu}{2}\cdot\frac{\mathbf{1}^\top V_c^{-1}B_c}{1+B_c^\top V_c^{-1}B_c}.\tag{6.2}$$

Multiplying (6.1) by $\mathbf{1}^\top$ and using (6.2):

$$1 = \frac{\nu}{2}\left[\mathbf{1}^\top V_c^{-1}\mathbf{1} - \frac{\left(\mathbf{1}^\top V_c^{-1}B_c\right)^2}{1 + B_c^\top V_c^{-1}B_c}\right] \quad\Longrightarrow\quad \frac{\nu}{2} = \frac{1 + B_c^\top V_c^{-1}B_c}{\left(1 + B_c^\top V_c^{-1}B_c\right)\left(\mathbf{1}^\top V_c^{-1}\mathbf{1}\right) - \left(\mathbf{1}^\top V_c^{-1}B_c\right)^2}, \tag{6.3}$$

the denominator being strictly positive because, by the Cauchy–Schwarz inequality in the inner product $\langle u, v\rangle = u^\top V_c^{-1} v$,

$$\left(\mathbf{1}^\top V_c^{-1}B_c\right)^2 \le \left(\mathbf{1}^\top V_c^{-1}\mathbf{1}\right)\left(B_c^\top V_c^{-1}B_c\right) < \left(1 + B_c^\top V_c^{-1}B_c\right)\left(\mathbf{1}^\top V_c^{-1}\mathbf{1}\right). \tag{6.4}$$

Substituting (6.2)–(6.3) into (6.1) gives the stated abstract formula. The optimal value: from the stationarity equation, $V_c\,\omega = \frac{\nu}{2}\mathbf{1} - aB_c$, so $\omega^\top V_c\,\omega = \frac{\nu}{2}\,\mathbf{1}^\top\omega - a\,\omega^\top B_c = \frac{\nu}{2} - a^2$ and

$$M^* = a^2 + \omega^\top V_c\,\omega = \frac{\nu}{2}. \tag{6.5}$$

Explicit forms. With $V_c^{-1} = (\gamma^2 S_0)^{-1}\operatorname{diag}(c_1^{-1}, \dots, c_m^{-1})$ and $B_c = \beta(d_j^\rho)_j$:

$$\mathbf{1}^\top V_c^{-1}\mathbf{1} = \frac{1}{\gamma^2}, \qquad \mathbf{1}^\top V_c^{-1}B_c = \frac{\beta S_\rho}{\gamma^2 S_0}, \qquad B_c^\top V_c^{-1}B_c = \frac{\beta^2 S_{2\rho}}{\gamma^2 S_0}.$$

Then $1 + B_c^\top V_c^{-1}B_c = \dfrac{\gamma^2 S_0 + \beta^2 S_{2\rho}}{\gamma^2 S_0}$, and the j-th coordinate of the numerator of $\omega^{(\mathrm{AMSE})}$ is

$$\frac{\gamma^2 S_0 + \beta^2 S_{2\rho}}{\gamma^2 S_0}\cdot\frac{c_j^{-1}}{\gamma^2 S_0} - \frac{\beta S_\rho}{\gamma^2 S_0}\cdot\frac{\beta\, c_j^{-1} d_j^\rho}{\gamma^2 S_0} = \frac{c_j^{-1}}{\gamma^4 S_0^2}\left[\gamma^2 S_0 + \beta^2 S_{2\rho} - \beta^2 S_\rho d_j^\rho\right],$$

while the denominator equals

$$\frac{\gamma^2 S_0 + \beta^2 S_{2\rho}}{\gamma^2 S_0}\cdot\frac{1}{\gamma^2} - \frac{\beta^2 S_\rho^2}{\gamma^4 S_0^2} = \frac{\left(\gamma^2 S_0 + \beta^2 S_{2\rho}\right)S_0 - \beta^2 S_\rho^2}{\gamma^4 S_0^2} = \frac{\Delta}{\gamma^4 S_0^2}.$$

Dividing yields $\omega_j^{(\mathrm{AMSE})} = c_j^{-1}\left[\gamma^2 S_0 + \beta^2 S_{2\rho} - \beta^2 S_\rho d_j^\rho\right]/\Delta$; one checks $\sum_j \omega_j^{(\mathrm{AMSE})} = \left[S_0(\gamma^2 S_0 + \beta^2 S_{2\rho}) - \beta^2 S_\rho^2\right]/\Delta = 1$, and $\Delta > 0$ restates (6.4) (or directly: $S_0 S_{2\rho} \ge S_\rho^2$ by Cauchy–Schwarz, so $\Delta \ge \gamma^2 S_0^2 > 0$). Finally, from (6.3) and (6.5),

$$M^* = \frac{\nu}{2} = \frac{(\gamma^2 S_0 + \beta^2 S_{2\rho})/(\gamma^2 S_0)}{\Delta/(\gamma^4 S_0^2)} = \frac{\gamma^2 S_0\left(\gamma^2 S_0 + \beta^2 S_{2\rho}\right)}{\Delta}.$$

(Sanity check: if $\lambda = 0$ then $\beta = 0$, $M^* = \gamma^2 = M(\omega^{(\mathrm{Var})})$ and $\omega^{(\mathrm{AMSE})} = \omega^{(\mathrm{Var})}$.)

(ii) Since the d_j are not all equal to 1, and $\sum_j \pi_j d_j = 1$ (Lemma 1(ii)), the d_j — hence the d_j^ρ, as $\rho \neq 0$ — are non-constant in j. Two strict inequalities follow:

- **Strict Cauchy–Schwarz:** $S_0 S_{2\rho} - S_\rho^2 > 0$, because equality in $\left(\sum_j c_j^{-1} d_j^\rho\right)^2 \le \left(\sum_j c_j^{-1}\right)\left(\sum_j c_j^{-1} d_j^{2\rho}\right)$ would force d_j^ρ constant.
- **Strict Jensen** (as in Theorem 5(ii)): $S_\rho/S_0 = \sum_j \pi_j d_j^\rho > \left(\sum_j \pi_j d_j\right)^\rho = 1$, hence $S_\rho^2 - S_0^2 > 0$.

Thus $\lambda_0 \in (0, \infty)$ is well defined. Now compare M^* with $M^{\mathrm{or}} = \beta^2 + \gamma^2$, using $\Delta > 0$:

$$M^* \ge M^{\mathrm{or}} \iff \gamma^2 S_0\left(\gamma^2 S_0 + \beta^2 S_{2\rho}\right) \ge \left(\gamma^2 + \beta^2\right)\Delta \iff \gamma^2\left[S_0\left(\gamma^2 S_0 + \beta^2 S_{2\rho}\right) - \Delta\right] \ge \beta^2\Delta.$$

But $S_0\left(\gamma^2 S_0 + \beta^2 S_{2\rho}\right) - \Delta = \beta^2 S_\rho^2$ by definition of Δ, so the comparison reads $\gamma^2\beta^2 S_\rho^2 \ge \beta^2\Delta$. If $\beta = 0$ (i.e. $\lambda = 0$) this holds with equality, consistently with $0 = |\lambda| \le \lambda_0$. If $\beta \neq 0$, divide by $\beta^2 > 0$:

$$M^* \ge M^{\mathrm{or}} \iff \gamma^2 S_\rho^2 \ge \Delta = \gamma^2 S_0^2 + \beta^2\left(S_0 S_{2\rho} - S_\rho^2\right) \iff \beta^2 \le \gamma^2\,\frac{S_\rho^2 - S_0^2}{S_0 S_{2\rho} - S_\rho^2} \iff |\lambda| \le \lambda_0,$$

where the last step uses $\beta^2=\lambda^2/(1-\rho)^2$. ■

Remark 3 (interpretation). When $|\lambda|\le\lambda_0$ (small bias), the price of distribution under unbalanced designs is a (typically small) AMSE inflation even with optimal weights; when $|\lambda|>\lambda_0$ (large bias), the heterogeneity of the machine-level biases $\lambda_j\propto d_j^\rho$ becomes an *asset*: weighting can partially cancel bias against bias, and the feasible distributed estimator strictly improves on the infeasible benchmark — for expectiles exactly as DPS found for the tail index and extreme quantiles. In the balanced case $d_j\equiv 1$ one has $S_\alpha=S_0$ for every α, so all machine-level biases coincide with the benchmark bias; then $\Delta=(\gamma^2S_0+\beta^2S_0)S_0-\beta^2S_0^2=\gamma^2S_0^2$ and $M^*=\gamma^2S_0\cdot S_0(\gamma^2+\beta^2)/(\gamma^2S_0^2)=\gamma^2+\beta^2=M^{\mathrm{or}}$: with balanced designs nothing can beat (or lose to) the benchmark at first order, and the threshold phenomenon disappears.

6.4 Data-driven AMSE weights from the augmented protocol

The weights $\omega^{(\mathrm{AMSE})}$ involve (γ,β',ρ)-type quantities; following DPS, Section 3.2, the protocol is augmented so that machine j also transmits second-order estimates $(\widehat\beta_j,\hat\rho_j)$, and one assumes the Hall–Welsh-type representation

$$\textbf{(A5)}\qquad A(t)=\gamma\,\beta'\,t^\rho\quad\text{for some }\beta'\neq 0,\qquad \widehat\beta_j\xrightarrow{P}\beta',\qquad (\hat\rho_j-\rho)\log n_j=o_P(1),\quad 1\le j\le m.$$

(These are the consistency requirements DPS impose; they are met, e.g., by the estimators of Gomes and Martins implemented in `evt0::mop` .) The central machine forms

$$\widehat\beta_n=\sum_{j=1}^m\frac{n_j}{n}\widehat\beta_j,\qquad \hat\rho_n=\sum_{j=1}^m\frac{n_j}{n}\hat\rho_j,\qquad \widetilde\gamma_n=\widehat\gamma_n\big(\widetilde\omega_n^{(\mathrm{Var})}\big),$$

$$\widetilde B_{c,j}=\frac{\sqrt{k}\,\widetilde\gamma_n\,\widehat\beta_n}{1-\hat\rho_n}\Big(\frac{n_j}{k_j}\Big)^{\hat\rho_n},\qquad \widetilde V_c=k\,\widetilde\gamma_n^2\,\mathrm{diag}\Big(\frac{1}{k_1},\dots,\frac{1}{k_m}\Big),$$

and plugs $(\widetilde B_c,\widetilde V_c)$ into the abstract formula of Theorem 6(i) to obtain $\widetilde\omega_n^{(\mathrm{AMSE})}$. Everything is computable from $\{T_j,(\widehat\beta_j,\hat\rho_j)\}_{j\le m}$.

> **Corollary 7 (feasible AMSE-optimal distributed estimator).** *Assume (A1)–(A5). Then* $\widetilde\omega_n^{(\mathrm{AMSE})}\xrightarrow{P}\omega^{(\mathrm{AMSE})}$ *and*
>
> $$\frac{\sqrt{k}}{D_n}\Big(\frac{\hat\xi_n^*\big(1-p\,|\,\widetilde\omega_n^{(\mathrm{AMSE})}\big)}{\xi_{1-p}}-1\Big)\xrightarrow{d}\mathcal N\Big(\big(\omega^{(\mathrm{AMSE})}\big)^\top B_c,\ \big(\omega^{(\mathrm{AMSE})}\big)^\top V_c\,\omega^{(\mathrm{AMSE})}\Big),$$
>
> *whose (squared bias + variance) equals the optimal value M^* of Theorem 6.*

Proof. *Consistency of the plug-ins.* First, $\widetilde V_c\to V_c$: indeed $[\widetilde V_c]_{jj}=\widetilde\gamma_n^2\,k/k_j\xrightarrow{P}\gamma^2\kappa_j=\gamma^2S_0c_j=[V_c]_{jj}$, by Lemma 1(i) and $\widetilde\gamma_n\xrightarrow{P}\gamma$ (Theorem 2's Step 3 with $\omega=\omega^{(\mathrm{Var})}$, or simply Proposition A and Lemma 1).

Second, $\widetilde B_c\xrightarrow{P}B_c$. Under (A5), $\sqrt{k}A(n_j/k_j)=\sqrt{k}\,\gamma\beta'(n_j/k_j)^\rho\to d_j^\rho\lambda$ by Lemma 1(iii), so $B_{c,j}=\beta d_j^\rho=\lim_n\sqrt{k}\,\gamma\beta'(n_j/k_j)^\rho/(1-\rho)$ and it suffices to show

$$R_{n,j}:=\frac{\widetilde B_{c,j}}{\sqrt{k}\,\gamma\beta'(n_j/k_j)^\rho/(1-\rho)}=\frac{\widetilde\gamma_n\widehat\beta_n}{\gamma\beta'}\cdot\frac{1-\rho}{1-\hat\rho_n}\cdot\Big(\frac{n_j}{k_j}\Big)^{\hat\rho_n-\rho}\xrightarrow{P}1.$$

The first two factors tend to 1 in probability since $\widetilde\gamma_n\xrightarrow{P}\gamma$, $\widehat\beta_n\xrightarrow{P}\beta'$ (convex combination of consistent estimators) and $\hat\rho_n\xrightarrow{P}\rho$. For the third, $\big(\frac{n_j}{k_j}\big)^{\hat\rho_n-\rho}=\exp\big[(\hat\rho_n-\rho)\log(n_j/k_j)\big]$ and $0\le\log(n_j/k_j)\le\log n_j\le\log n$ eventually, while

$$(\hat\rho_n-\rho)\log n=\sum_{j=1}^m\frac{n_j}{n}\,(\hat\rho_j-\rho)\log n_j\cdot\frac{\log n}{\log n_j}=o_P(1),$$

because each $(\hat\rho_j-\rho)\log n_j=o_P(1)$ by (A5) and $\log n/\log n_j\to 1$ (indeed $\log n-\log n_j\to\log(b_j\sum_i b_i^{-1})$ is bounded while $\log n_j\to\infty$). Hence $R_{n,j}\xrightarrow{P}1$ and $\widetilde B_c\xrightarrow{P}B_c$.

Consistency of the weights. The map $(B,V)\mapsto\omega(B,V)$ given by the abstract formula of Theorem 6(i) is continuous at (B_c,V_c): its denominator is $\big(1+B^\top V^{-1}B\big)\big(\mathbf{1}^\top V^{-1}\mathbf{1}\big)-\big(\mathbf{1}^\top V^{-1}B\big)^2$, which at (B_c,V_c) is strictly positive by (6.4) — note this holds for *any* B_c, including $B_c=0$, so no extra condition on λ or the d_j is needed. By the continuous mapping theorem, $\widetilde{\omega}_n^{(\mathrm{AMSE})}\xrightarrow{P}\omega^{(\mathrm{AMSE})}$. Also $\mathbf{1}^\top\widetilde{\omega}_n^{(\mathrm{AMSE})}=1$ identically, by construction of the formula (it is normalized for every (B,V) in its domain).

Conclusion. Apply Theorem 2 with $\widehat{\omega}_n=\widetilde{\omega}_n^{(\mathrm{AMSE})}$ and $\omega=\omega^{(\mathrm{AMSE})}$. ■

6.5 Bias-reduced pooled estimator

The bias $\omega^\top B_c$ of the limit law is *known up to consistently estimable quantities* once the augmented protocol is in force; since the error analysis lives on the $\log$ scale, the natural correction is multiplicative.

> **Corollary 8 (bias-reduced distributed extreme expectile estimator).** *Assume (A1)–(A5), and let $\widehat{\omega}_n\xrightarrow{P}\omega\in\mathcal{W}$ with $\mathbf{1}^\top\widehat{\omega}_n=1$. Define*
>
> $$\overline{\xi}_n^{\,*}(1-p\,|\,\widehat{\omega}_n):=\hat{\xi}_n^{*}(1-p\,|\,\widehat{\omega}_n)\ \exp\Big(-\frac{D_n}{\sqrt{k}}\,\widehat{\omega}_n^\top\widetilde{B}_c\Big).$$
>
> *Then*
>
> $$\frac{\sqrt{k}}{D_n}\Big(\frac{\overline{\xi}_n^{\,*}(1-p\,|\,\widehat{\omega}_n)}{\xi_{1-p}}-1\Big)\xrightarrow{d}\mathcal{N}\Big(0,\ \gamma^2S_0\sum_j c_j\omega_j^2\Big).$$
>
> *In particular, with $\widehat{\omega}_n=\widetilde{\omega}_n^{(\mathrm{Var})}$ the limit is $\mathcal{N}(0,\gamma^2)$: the distributed estimator attains the benchmark variance with no first-order bias.*

Proof. On the $\log$ scale,

$$\frac{\sqrt{k}}{D_n}\log\frac{\overline{\xi}_n^{\,*}(1-p|\widehat{\omega}_n)}{\xi_{1-p}}=\frac{\sqrt{k}}{D_n}\log\frac{\hat{\xi}_n^{*}(1-p|\widehat{\omega}_n)}{\xi_{1-p}}-\widehat{\omega}_n^\top\widetilde{B}_c.$$

By Theorem 2 (proof, Steps 1–2), the first term equals $\sqrt{k}\big(\widehat{\gamma}_n(\omega)-\gamma\big)+o_P(1)\xrightarrow{d}\mathcal{N}\big(\omega^\top B_c,\ \gamma^2S_0\sum_j c_j\omega_j^2\big)$; by Corollary 7's proof, $\widetilde{B}_c\xrightarrow{P}B_c$, so $\widehat{\omega}_n^\top\widetilde{B}_c\xrightarrow{P}\omega^\top B_c$. Slutsky's lemma gives the centred limit for the $\log$ ratio; the correction factor $\exp\big(-(D_n/\sqrt{k})\widehat{\omega}_n^\top\widetilde{B}_c\big)\xrightarrow{P}1$ (as $D_n/\sqrt{k}\to0$), so consistency in relative terms is preserved and the $\log\mapsto$ relative-error conversion of Theorem 1's proof applies. ■

7. Asymptotic confidence intervals

> **Corollary 9 (log-scale confidence intervals computable at the central machine).** *Assume (A1)–(A4), let $\widehat{\omega}_n\xrightarrow{P}\omega\in\mathcal{W}$ with $\mathbf{1}^\top\widehat{\omega}_n=1$, and define the variance estimator*
>
> $$\widehat{\sigma}_n^2:=\widehat{\gamma}_n\big(\widetilde{\omega}_n^{(\mathrm{Var})}\big)^2\ k\sum_{j=1}^m\frac{\widehat{\omega}_{n,j}^2}{k_j}\qquad\Big(\xrightarrow{P}\ \gamma^2S_0\sum_j c_j\omega_j^2\Big).$$
>
> *Let $z_{1-\alpha/2}$ be the standard normal quantile and set, for a generic point estimator $\check{\xi}_n$,*
>
> $$\mathcal{I}_{n,1-\alpha}\big(\check{\xi}_n\big):=\Big[\check{\xi}_n\ \exp\Big(-z_{1-\alpha/2}\,\frac{D_n}{\sqrt{k}}\,\widehat{\sigma}_n\Big),\ \ \check{\xi}_n\ \exp\Big(z_{1-\alpha/2}\,\frac{D_n}{\sqrt{k}}\,\widehat{\sigma}_n\Big)\Big].$$
>
> *Then:*
>
> *(i) if $\lambda=0$, $\mathbb{P}\big(\xi_{1-p}\in\mathcal{I}_{n,1-\alpha}(\hat{\xi}_n^{*}(1-p|\widehat{\omega}_n))\big)\to1-\alpha$;*
>
> *(ii) under (A5) and with arbitrary $\lambda\in\mathbb{R}$, $\mathbb{P}\big(\xi_{1-p}\in\mathcal{I}_{n,1-\alpha}(\overline{\xi}_n^{\,*}(1-p|\widehat{\omega}_n))\big)\to1-\alpha$.*

> *With $\widehat{\omega}_n = \widetilde{\omega}_n^{(\mathrm{Var})}$ one has $k\sum_j \widehat{\omega}_{n,j}^2/k_j = 1$ exactly, so $\widehat{\sigma}_n = \widehat{\gamma}_n(\widetilde{\omega}_n^{(\mathrm{Var})})$ and the interval has half-width $z_{1-\alpha/2}\,\widehat{\gamma}_n(\widetilde{\omega}_n^{(\mathrm{Var})})\ \log(k/(np))/\sqrt{k}$ on the $\log$ scale — formally identical to the one-sample QB interval, with k the combined effective sample size.*

Proof. *Variance consistency.* $k\sum_j \widehat{\omega}_{n,j}^2/k_j = \sum_j \widehat{\omega}_{n,j}^2\,(k/k_j) \xrightarrow{P} \sum_j \omega_j^2\kappa_j = S_0\sum_j c_j\omega_j^2$ by Lemma 1(i), and $\widehat{\gamma}_n(\widetilde{\omega}_n^{(\mathrm{Var})}) \xrightarrow{P} \gamma$; hence the stated limit, which is the variance in Theorem 2 (and Corollary 8).

(i) If $\lambda = 0$, the bias $\beta\sum_j \omega_j d_j^\rho$ vanishes, so by Theorem 2 and Slutsky,

$$Z_n := \frac{\sqrt{k}}{D_n\,\widehat{\sigma}_n}\Big(\frac{\hat{\xi}_n^*(1-p|\widehat{\omega}_n)}{\xi_{1-p}} - 1\Big) \xrightarrow{d} \mathcal{N}(0,1).$$

Now $\xi_{1-p} \in \mathcal{I}_{n,1-\alpha}$ iff $\big|\log\big(\hat{\xi}_n^*/\xi_{1-p}\big)\big| \le z_{1-\alpha/2}(D_n/\sqrt{k})\widehat{\sigma}_n$; since $\log(\hat{\xi}_n^*/\xi_{1-p}) = \big(\hat{\xi}_n^*/\xi_{1-p} - 1\big)(1 + o_P(1))$ (consistency in relative terms, proof of Theorem 2), this event differs from $\{|Z_n| \le z_{1-\alpha/2}(1 + o_P(1))\}$ by an asymptotically negligible amount, and its probability tends to $\mathbb{P}(|\mathcal{N}(0,1)| \le z_{1-\alpha/2}) = 1 - \alpha$.

(ii) Identical argument with Corollary 8 in place of Theorem 2 (the limit is centred for any λ).

Last claim. With $\widehat{\omega}_{n,j} = k_j/k$: $k\sum_j (k_j/k)^2/k_j = \sum_j k_j/k = 1$. ■

8. Equivalent designs, and why not the arithmetic mean

The central machine could organize the same transmitted information differently. Two natural competitors are

$$\textbf{(E2)}\quad \hat{\xi}_n^{(2)}(1-p\,|\,\widehat{\omega}_n) := \varphi\big(\widehat{\gamma}_n(\widehat{\omega}_n)\big)\prod_{j=1}^m \big[\hat{q}_j^*(1-p)\big]^{\widehat{\omega}_{n,j}}$$

(pool the Weissman quantile estimators geometrically as in DPS, then apply the Bellini factor with the *pooled* Hill estimator), and

$$\textbf{(E3)}\quad \hat{\xi}_n^{(3)}(1-p\,|\,\widehat{\omega}_n) := \varphi\big(\widehat{\gamma}_n(\widehat{\omega}_n)\big)\prod_{j=1}^m \Big[\Big(\frac{k_j}{n_j p}\Big)^{\widehat{\gamma}_n(\widehat{\omega}_n)}\widehat{Q}_j\Big]^{\widehat{\omega}_{n,j}}$$

(use the pooled Hill estimator everywhere, including inside each machine's extrapolation — the analogue of DPS's "$\hat{q}_j^*(1-p|k_j,\omega)$" design).

> **Proposition 10 (asymptotic equivalence of designs).** *Assume (A1)–(A4) and let $\widehat{\omega}_n \xrightarrow{P} \omega \in \mathcal{W}$, $\mathbf{1}^\top\widehat{\omega}_n = 1$. Then, for $i \in \{2,3\}$,*
>
> $$\frac{\sqrt{k}}{D_n}\left(\frac{\hat{\xi}_n^{(i)}(1-p\,|\,\widehat{\omega}_n)}{\hat{\xi}_n^*(1-p\,|\,\widehat{\omega}_n)} - 1\right) = o_P(1),$$
>
> *so Theorems 2, 5, 6 and Corollaries 4, 7, 8, 9 hold verbatim for $\hat{\xi}_n^{(2)}$ and $\hat{\xi}_n^{(3)}$.*

Proof. Write $k_{\min} := \min_j k_j$; by (A2), $k/k_{\min} = O(1)$.

(E2). On the $\log$ scale,

$$\log\hat{\xi}_n^{(2)} - \log\hat{\xi}_n^* = \log\varphi\big(\widehat{\gamma}_n(\widehat{\omega}_n)\big) - \sum_j \widehat{\omega}_{n,j}\log\varphi(\widehat{\gamma}_j).$$

A second-order Taylor expansion of $\log\varphi$ around γ (legitimate on an event of probability $\to 1$, where all of $\widehat{\gamma}_j, \widehat{\gamma}_n(\widehat{\omega}_n)$ lie in a fixed compact subinterval of $(0,1)$ on which $(\log\varphi)''$ is bounded) gives

$$\log\varphi(x) = \log\varphi(\gamma) + m(\gamma)(x-\gamma) + O\big((x-\gamma)^2\big)\quad \text{uniformly there,}$$

so, using $\sum_j \widehat{\omega}_{n,j} = 1$ and $\widehat{\gamma}_n(\widehat{\omega}_n) = \sum_j \widehat{\omega}_{n,j}\widehat{\gamma}_j$,

$$\log \hat{\xi}_n^{(2)} - \log \hat{\xi}_n^* = m(\gamma)\Big[\widehat{\gamma}_n(\widehat{\omega}_n) - \sum_j \widehat{\omega}_{n,j}\widehat{\gamma}_j\Big] + O_P\Big(\max_j(\widehat{\gamma}_j-\gamma)^2 + \big(\widehat{\gamma}_n(\widehat{\omega}_n)-\gamma\big)^2\Big) = 0 + O_P\big(k_{\min}^{-1}\big).$$

Multiplying by $\sqrt{k}/D_n$: $O_P\big(\sqrt{k}/(k_{\min}D_n)\big) = O_P\big((k/k_{\min})\,k^{-1/2}D_n^{-1}\big)\cdot O(1) = o_P(1)$.

(E3). There is the additional term

$$\sum_j \widehat{\omega}_{n,j}\big(\widehat{\gamma}_n(\widehat{\omega}_n)-\widehat{\gamma}_j\big)D_{n,j} = D_n\underbrace{\sum_j \widehat{\omega}_{n,j}\big(\widehat{\gamma}_n(\widehat{\omega}_n)-\widehat{\gamma}_j\big)}_{=\,\widehat{\gamma}_n(\widehat{\omega}_n)-\widehat{\gamma}_n(\widehat{\omega}_n)\,=\,0} + \sum_j \widehat{\omega}_{n,j}\big(\widehat{\gamma}_n(\widehat{\omega}_n)-\widehat{\gamma}_j\big)\,\delta_{n,j},$$

with $\delta_{n,j} := D_{n,j} - D_n \to -\log d_j$ bounded (Lemma 1(iv)). The first part vanishes *exactly* — this is the algebraic payoff of geometric pooling. The second is $O_P\big(k_{\min}^{-1/2}\big)$ since $\widehat{\gamma}_n(\widehat{\omega}_n)-\widehat{\gamma}_j = O_P(k_{\min}^{-1/2})$; multiplied by $\sqrt{k}/D_n$ it is $O_P\big(\sqrt{k/k_{\min}}/D_n\big) = o_P(1)$. Adding the (E2)-type Bellini-factor term, already shown to be negligible, completes the proof of the $\log$-scale equivalence; since both estimators are consistent in relative terms, the relative-error statement follows as before. ■

Remark 4 (one more variant). The "strength-borrowing" *marginal* estimator $\hat{\xi}_j^*(1-p\,|\,k_j,\omega) := \varphi(\widehat{\gamma}_n(\omega))\,d_{n,j}^{\widehat{\gamma}_n(\omega)}\widehat{Q}_j$, in which a single machine's extrapolation is driven by the pooled tail index, satisfies, by the very same decomposition as in Theorem 1 with $\widehat{\gamma}_j$ replaced by $\widehat{\gamma}_n(\omega)$ in terms T_1–T_2,

$$\frac{\sqrt{k}}{D_{n,j}}\Big(\frac{\hat{\xi}_j^*(1-p|k_j,\omega)}{\xi_{1-p}} - 1\Big) = \sqrt{k}\big(\widehat{\gamma}_n(\omega)-\gamma\big) + o_P(1):$$

each machine inherits the *pooled* rate $\sqrt{k}/D_n$ instead of its own $\sqrt{k_j}/D_{n,j}$ (the terms T_3–T_5 remain $O_P(k_j^{-1/2}) = O_P(k^{-1/2})$). This mirrors DPS, Section 2.3.

Remark 5 (arithmetic pooling). The naive convex combination $\check{\xi}_n(1-p|\omega) := \sum_j \omega_j\hat{\xi}_j^*(1-p)$ (with $\omega_j \geq 0$) satisfies the *identical* first-order theory: indeed, exactly,

$$\frac{\check{\xi}_n(1-p|\omega)}{\xi_{1-p}} - 1 = \sum_j \omega_j\Big(\frac{\hat{\xi}_j^*(1-p)}{\xi_{1-p}} - 1\Big),$$

and Theorem 1 plus the rescaling argument of Theorem 2 give the same limit. The preference for geometric pooling is therefore *not* first-order asymptotic but structural and second-order: (a) only geometric pooling is exactly linear in the transmitted Hill estimators on the $\log$ scale, which is what makes the exact cancellation in Proposition 10(E3) and the multiplicative bias correction of Corollary 8 possible; (b) log-scale inference is the recommended scale for extreme value statistics (Drees, 2003); (c) empirically, DPS (Section 5.1.1) find that geometric pooling of Weissman-type estimators outperforms arithmetic pooling by a wide margin in bias and MSE — the same mechanism (a multiplicative, heavily right-skewed estimator) operates verbatim here, with the additional skewed factor $\varphi(\widehat{\gamma}_j)$.

9. Summary of the procedure

Communication protocol (one-shot).

Machine j ($1 \leq j \leq m$): choose k_j; compute and transmit

$$T_j = \big(n_j,\ k_j,\ \widehat{\gamma}_j(k_j),\ X_{n_j-k_j:n_j,j}\big) \qquad \big[\text{augmented: also } (\widehat{\beta}_j,\hat{\rho}_j)\big].$$

Central machine:

1. **Weights.** Variance-optimal: $\widetilde{\omega}_n^{(\mathrm{Var})} = (k_1/k, \ldots, k_m/k)$. AMSE-optimal (augmented protocol): build $\widetilde{B}_c, \widetilde{V}_c$ as in Section 6.4 and plug into Theorem 6(i) to get $\widetilde{\omega}_n^{(\mathrm{AMSE})}$.
2. **Pooled tail index.** $\widehat{\gamma}_n(\widehat{\omega}_n) = \sum_j \widehat{\omega}_{n,j}\widehat{\gamma}_j$.
3. **Pooled extreme expectile.**

$$\hat{\xi}_n^*(1-p \,|\, \widehat{\omega}_n) = \prod_{j=1}^{m} \left[\left(\widehat{\gamma}_j^{-1} - 1\right)^{-\widehat{\gamma}_j} \left(\frac{k_j}{n_j p}\right)^{\widehat{\gamma}_j} X_{n_j - k_j : n_j, j} \right]^{\widehat{\omega}_{n,j}}$$

   (or, equivalently to first order, designs (E2)/(E3) of Proposition 10).
4. **Bias correction (optional, augmented protocol).** Multiply by $\exp\left(-(D_n/\sqrt{k})\,\widehat{\omega}_n^\top \widetilde{B}_c\right)$ (Corollary 8).
5. **Confidence interval.** $\check{\xi}_n \exp\left(\pm z_{1-\alpha/2}\,\widehat{\sigma}_n \log(k/(np))/\sqrt{k}\right)$ with $\widehat{\sigma}_n$ of Corollary 9; with variance-optimal weights, $\widehat{\sigma}_n = \widehat{\gamma}_n(\widetilde{\omega}_n^{(\mathrm{Var})})$.

Guarantees. Rate $\sqrt{k}/\log(k/(np))$ — the rate of the infeasible benchmark using all n observations with effective sample size k (Theorems 1–2); benchmark variance γ^2 attained by $\widetilde{\omega}_n^{(\mathrm{Var})}$, with full asymptotic equivalence to the benchmark under near-equal sample fractions (Theorem 5); AMSE-optimality with the explicit threshold λ_0 beyond which the distributed estimator strictly *beats* the benchmark (Theorem 6); valid bias-corrected Gaussian inference at the central machine (Corollaries 8–9).

10. Discussion and possible extensions

(a) Why only the Hill bias survives at the extreme level. At an *intermediate* expectile level, QB estimation carries three first-order bias sources: the Hill bias, the second-order quantile bias, and the expectile–quantile gap bias (the λ_1- and λ_2-terms of Corollary 3.4 in Davison, Padoan and Stupfler, 2023). At the *extreme* level, all error sources of size $O_P(k_j^{-1/2})$ are divided by the diverging extrapolation factor $D_{n,j}$; only the term $(\widehat{\gamma}_j - \gamma)D_{n,j}$ keeps its size. The whole geometry of the problem thus collapses onto the pooled Hill estimator (Remark 2), which is what makes the DPS weighting theory portable. Conditions (A4) and $\rho < 0$ are exactly what is needed to certify that the discarded terms are $O_P(k_j^{-1/2})$ and not larger.

(b) LAWS is infeasible here — but nothing is lost at first order. Asymmetric-least-squares (LAWS) estimation of ξ_τ at intermediate levels requires the raw subsamples, which the protocol forbids. One could augment the protocol by letting machine j also transmit its intermediate LAWS expectile $\tilde{\xi}_{\tau_j,j}$ and pool $\tilde{\xi}_{\tau_j,j}\, d_{n,j}^{\widehat{\gamma}_j}$ geometrically; under the additional conditions of Theorem 3.5 of Davison, Padoan and Stupfler (2023) (notably $\gamma < 1/2$), the same limit theory would result, because LAWS-based and QB-based *extrapolating* estimators are asymptotically equivalent at extreme levels (Davison, Padoan and Stupfler, 2023, Theorem 3.5; Padoan and Stupfler, 2022, Theorem 2.4). The QB route adopted here is thus both the only one compatible with the DPS heuristics *and* costless in first-order efficiency — and it works on all of $\gamma \in (0,1)$ rather than $\gamma \in (0, 1/2)$.

(c) Composite levels. A popular device selects the expectile level $\tau_n'(p)$ with $\xi_{\tau_n'(p)} = q_{1-p}$, estimated by $1 - \widehat{\tau}_n'(p) = p\,\widehat{\gamma}_n/(1-\widehat{\gamma}_n)$. In the pure QB construction this device *collapses algebraically*: replacing $1-p$ by $1-\widehat{\tau}_n'(p)$ in $\hat{\xi}_n^*$ reproduces exactly the pooled Weissman quantile estimator of DPS, since $\varphi(\widehat{\gamma})\left[(\widehat{\gamma}^{-1}-1)^{-1}\right]^{-\widehat{\gamma}} = 1$. Composite estimation is therefore already covered by DPS, Corollary 9, and adds nothing new in the QB world.

(d) Choice of p, k_j and practical remarks. The theory allows np bounded (even $np \to 0$), the regime of typical interest, e.g. $p = 1/n$. The conditions on k_j are the usual one-sample ones imposed machine-by-machine, plus proportionality; the variance-optimal weights k_j/k require no extra estimation, while AMSE weights need the augmented protocol. The interval of Corollary 9 is asymmetric around the point estimate on the natural scale — appropriate for a right-skewed extrapolating estimator.

(e) Extensions.

1. *Serial dependence within machines.* Replacing Propositions A–B by their time-series versions under mixing and a tail dependence condition (Davison, Padoan and Stupfler, 2023, Sections 2 and 6; DPS, Section 4.1) changes only the variance constants ($\gamma^2 \rightsquigarrow \gamma^2(1+2\sum_{t\geq 1} R(t))$ -type factors per machine); the architecture of Theorems 1–2 (the reduction to the pooled tail index estimator) is untouched. Optimal weights then involve the machine-level long-run variances, which machines would additionally transmit.
2. *Growing number of machines.* DPS, Section 3.3, develop $m = m_n \to \infty$ for the tail index and quantiles; the reduction principle of Theorem 2 suggests the expectile analogue holds under their conditions plus uniform-in-j versions of (A3)–(A4); the bookkeeping of the $o_P(1)$ terms (now m_n of them) would require uniform bounds in Theorem 1, e.g. via the tail empirical process arguments of Chen, Li and Zhou (2021).

3. *Heterogeneous distributions.* DPS's general Sections 2.2–2.3 allow different marginals with a common γ. For expectiles, machine-specific proportionality constants $\varphi(\gamma)$ would coincide but the targets $\xi_{1-p,j}$ would differ through scale; a common target must then be defined (e.g. the expectile of a designated reference machine), and the bias bookkeeping of T_5 becomes machine-specific. This is a natural thesis follow-up.

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